THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND TWO MARK SCHEME

Sponsored by



Jane Street

1 Group Round

A sheet with the answers will be provided after the round ends, you can just check it. All or nothing - either they got it right, or they didn't. If you're unsure of the correctness of an answer, leave it and inform the organisers so they can grade it. The total score at the end is

$$6 \times (\#\text{ of correct answers}).$$

For the rest of the round, you don't need to be doing anything. Chill, relax, try to solve some integrals, do an example sheet etc.

Answers:

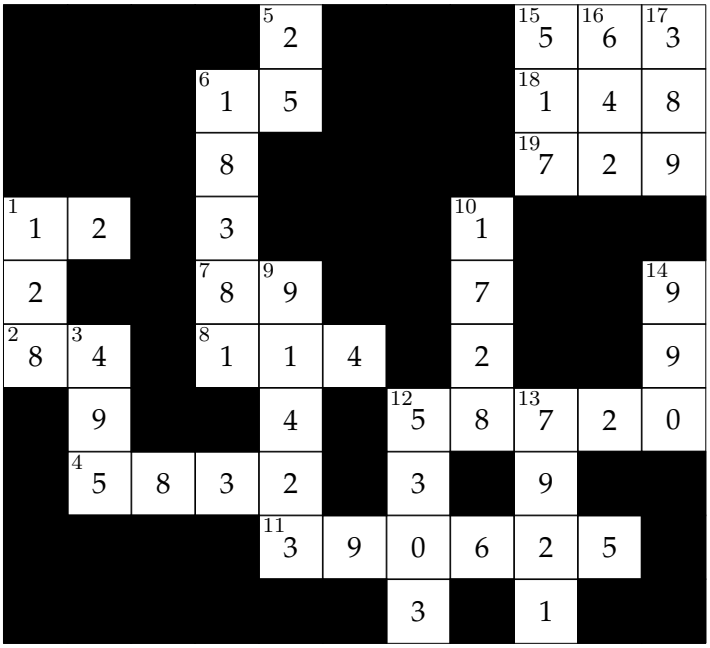
1. $\frac{1}{2}$
2. $\frac{(\ln(1-e^{-x}))^2}{2} + C$ or $\frac{(x-\ln(e^x-1))^2}{2} + C$.
3. π
4. $\frac{\pi}{2}\ln(1+\sqrt{2})$
5. $\frac{\ln(2)}{2} + \frac{1}{4}$
6. $\frac{3e^2}{4} - \frac{1}{4}$
7. $\frac{\pi(e-1)}{2}$
8. $\frac{\ln(1+\sqrt{2})}{2} - \frac{\pi}{4}$ or $\frac{\sinh^{-1}(1)}{2} - \frac{\pi}{4}$
9. $\frac{\pi}{2}\coth(\pi) - \frac{1}{2}$
10. $\frac{1}{2}(\ln(2))^2$

2 Crossnumber Round

This is the most involved round to grade. There is a master copy of the crossnumber grid to be placed between the half-teams/in front of you. A half-team will say they want to submit an answer, and likely point at or reference this master copy. Then they will either provide one digit at a time or several at once. If it's one at a time, grade each digit one at a time, if it's all at once, grade all of them (no chance to take back their digits).

When one of the half-teams submits a digit, write down the correct digit on the master copy, regardless of whether the submitted digit was right or wrong. Put a little tick next to it if it was right and a little cross next to it if it was wrong. The total score at the end is the number of ticks.

The completed crossnumber is the following:



Relay Round

The relay round is marked one problem at a time. 3 points for being correct on the first attempt, 1 point for being correct on any later attempt. They can also skip a question, in which case they get 0 points and they can't come back to it. Once they get the current problem right or skip it, give them the next problem. If they get 10-14 problems right (regardless of how many attempts they needed), they get an extra 5 points. If they get 15 or more problems right, they get an extra 10 points. Extra points are not additive. The total score at the end is

Answers:

1. $\dfrac{\pi^2 - 8}{4}$
2. $\ln(2)$
3. $\dfrac{1}{2}$
4. $2G - \dfrac{\pi}{2}$
5. $\dfrac{\pi}{\sqrt{5}}$
6. 2
7. $\dfrac{1}{2}$
8. $\dfrac{\pi}{2\sqrt{2}}$
9. 1
10. $2\Gamma(-\alpha)\cos\left(\dfrac{\pi}{2}\alpha\right)|\theta|^{\alpha}$
11. $\dfrac{\pi}{4}$
12. $16\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}+C$
13. 0
14. $\dfrac{2025}{2}$
15. $\dfrac{\pi}{2}$
16. $2\sqrt{2}$
17. $\dfrac{5+\pi\coth(\pi)}{16}$
18. 1
19. $e\Gamma\left(\dfrac{3}{2}\right)$ or $\dfrac{e\sqrt{\pi}}{2}$
20. $x+\dfrac{\ln(x^5+1)}{5}+C$